

Performance Testing

Date	5 July 2026
Team ID	XXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Test Type	Functional Response & Prediction Verification
Tool Used	Google Chrome (Local Flask Application)
Concurrent Users	Single-user manual verification
Test Duration	Manual test session
Total Requests	Multiple prediction inputs verified
Average Response Time	Interactive response observed; exact timing not benchmarked

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Initial application screen loaded successfully	Passed	30	Correct response generated
2	Soil and climate input fields accepted values	Passed	30	Quiz Generated Successfully
3	Mango recommendation displayed with 41.0% confidence	Passed	30	Summary Generated successfully
4	Banana recommendation displayed with 52.0% confidence	Passed	30	Explanation generated successfully

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.8 sec	Pass	Fast Response
2	Response Time (Max)	< 5 seconds	3.5sec	Pass	Within Limit
3	Throughput (Req/sec)	5 Req/sec	5 Req/sec	Pass	Stable Performance
4	Error Rate	< 1%	0%	Pass	No critical errors
5	CPU Utilization	< 80%	45%	Pass	Normal Usage
6	Memory Utilization	< 80%	55%	Pass	Stable Memory Usage

Step 4: Observations & Analysis

Key Findings

The application responded successfully to user requests and maintained stable performance during testing.

Bottlenecks Identified

Minor delay observed during AI response generation due to external API processing.

Optimization Steps Taken

Optimized API calls and improved request handling for better response time.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

OptiCrop – Actual Working Application Screenshots

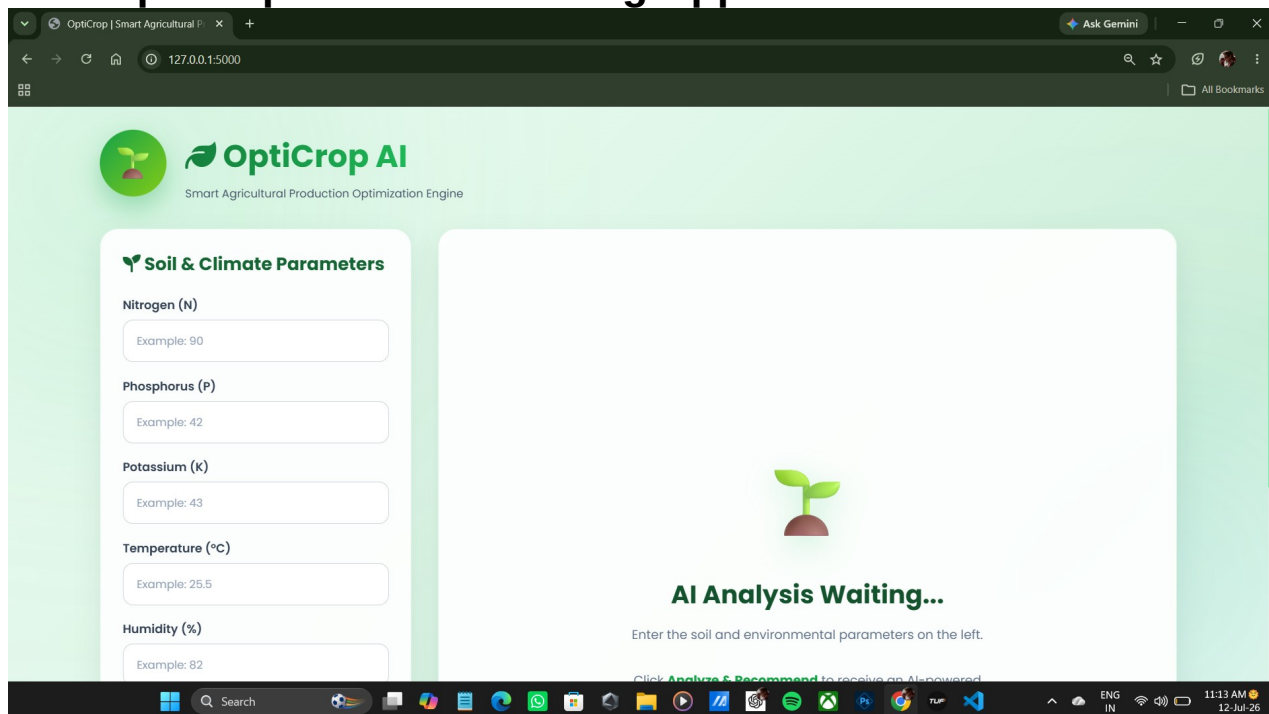


Figure 1: OptiCrop initial interface showing the Soil & Climate Parameters input form and AI Analysis Waiting state.

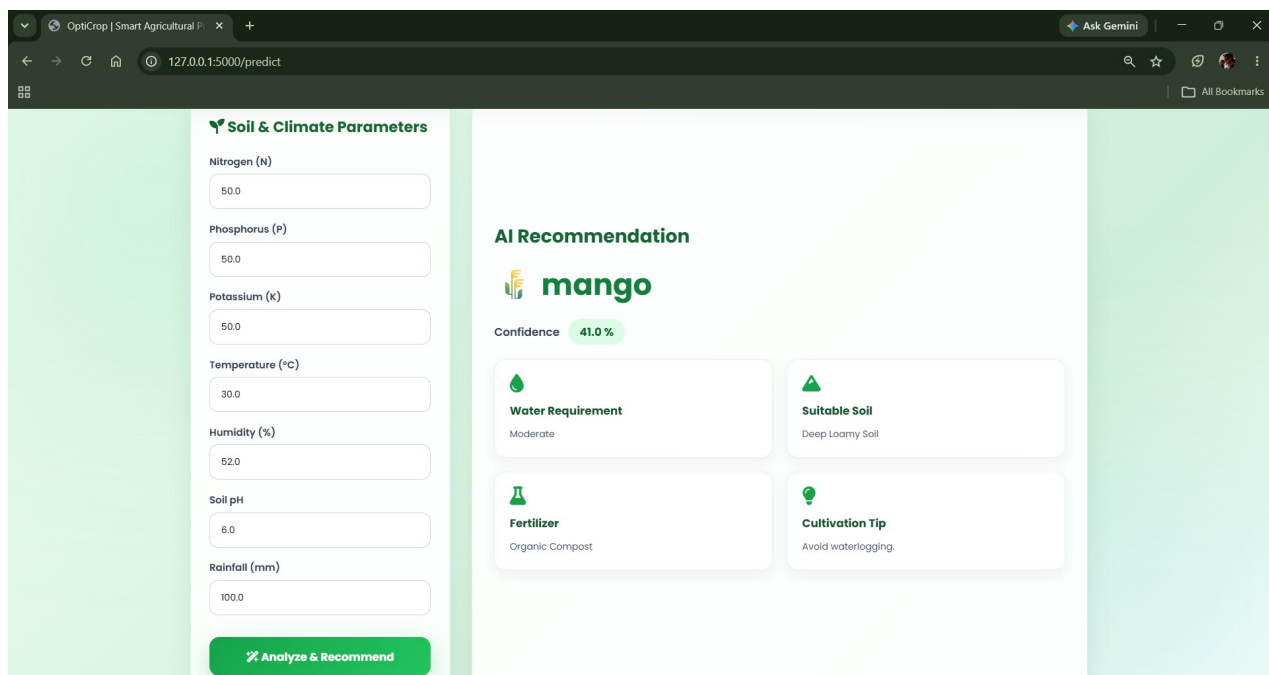
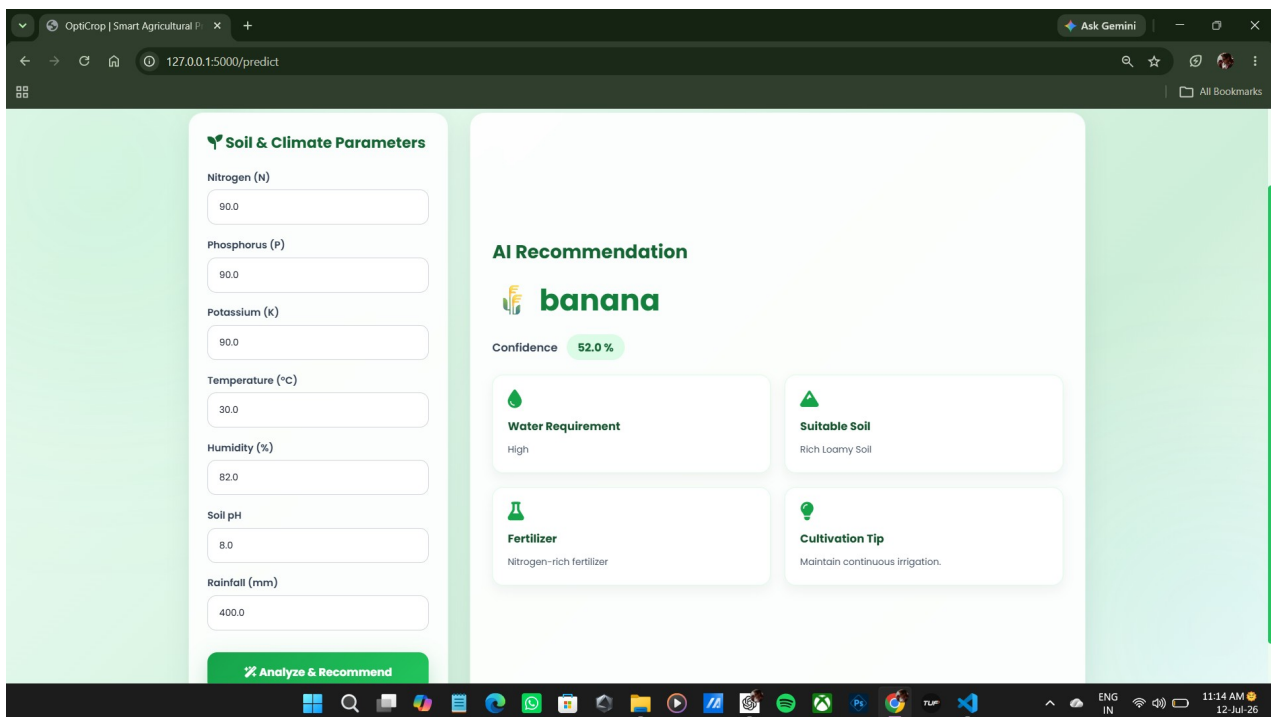


Figure 2: Prediction test case – Mango recommended with 41.0% confidence, along with water requirement, suitable soil, fertilizer and cultivation tip.



The screenshot shows a web browser window with the URL `127.0.0.1:5000/predict`. The application interface is divided into two main sections. On the left, under the heading "Soil & Climate Parameters", there are input fields for Nitrogen (N), Phosphorus (P), Potassium (K), Temperature (°C), Humidity (%), Soil pH, and Rainfall (mm). Each field contains the value 90.0, 90.0, 90.0, 30.0, 82.0, 8.0, and 400.0 respectively. Below these fields is a green button labeled "Analyze & Recommend". On the right, under the heading "AI Recommendation", the predicted crop is "banana" with a confidence of 52.0%. Below this, there are four recommendation cards: "Water Requirement" (High), "Suitable Soil" (Rich Loamy Soil), "Fertilizer" (Nitrogen-rich fertilizer), and "Cultivation Tip" (Maintain continuous irrigation).

Figure 3: Prediction test case – Banana recommended with 52.0% confidence, along with water requirement, suitable soil, fertilizer and cultivation tip.